

1 Spring and Mass

1.1 Problem and Dynamics:

$\hat{\lambda} = \frac{-\vec{\lambda}}{|\vec{\lambda}|} = \frac{-\vec{r}}{L}$ where $\vec{r} = x\hat{i} + y\hat{j}$ is position of mass, and $L = |\vec{r}|$

$T = K(L - L_0)$

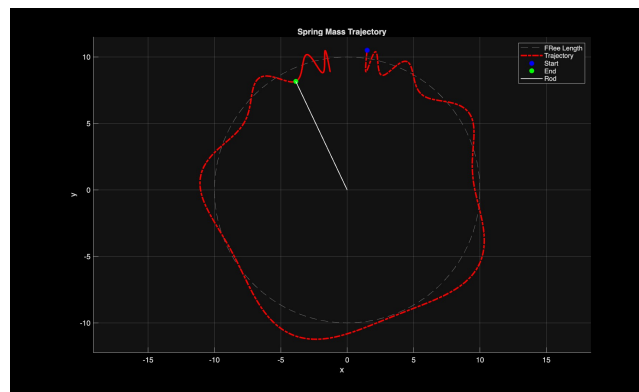
From FBD, $\sum \vec{F} = m\vec{a}$

$\Rightarrow \frac{T}{L}(-x\hat{i} - y\hat{j}) + mg(-\hat{j}) = m(\ddot{x}\hat{i} + \ddot{y}\hat{j})$

dotting with $\hat{i}$ and $\hat{j}$,

$$-\frac{T}{L}x = m\ddot{x} \quad \text{and} \quad -\frac{T}{L}y - mg = m\ddot{y}$$

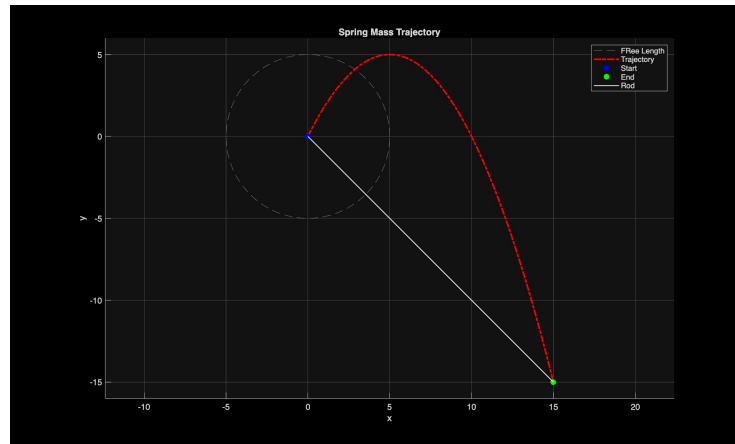
Dynamics Derivation



Motion with some constants and initial value

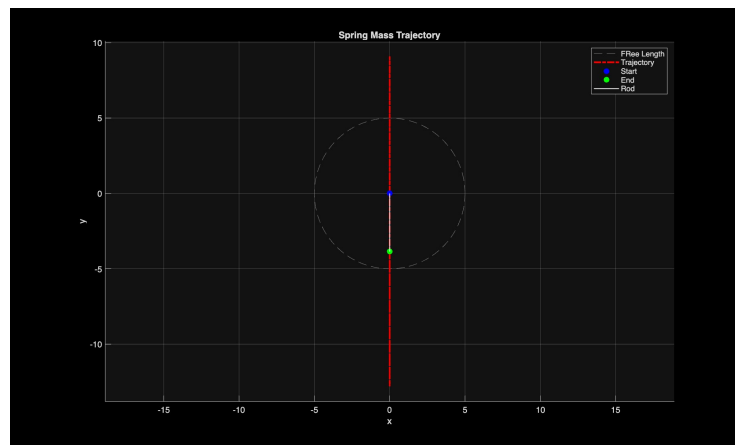
1.2 Checking the Solution

1.2.1 $k = 0$, all else arbitrary

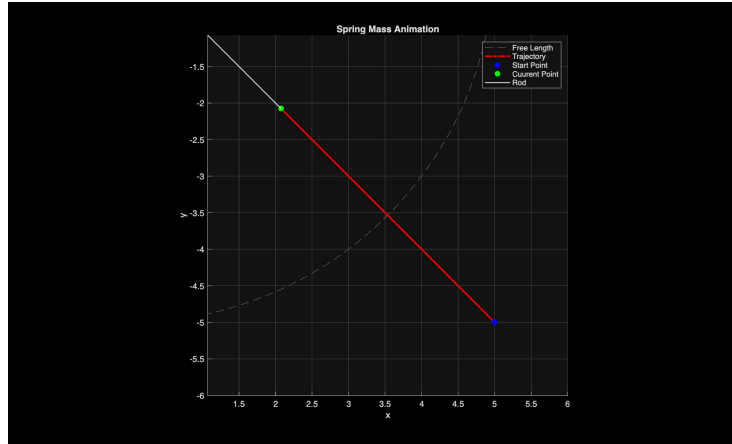


$K=0$

1.2.2 $x_0 = 0, v_{x0} = 0$



1.2.3 $g = 0, v_0 = 0$



COORDINATE SYSTEM STUFF

$\hat{e}_\lambda = -\sin\theta \hat{i} + \cos\theta \hat{j}$
 $\hat{e}_\theta = \cos\theta \hat{i} + \sin\theta \hat{j}$
 $\dot{\hat{e}}_\lambda = (-\cos\theta \hat{i} - \sin\theta \hat{j})\dot{\theta} = -\dot{\theta} \hat{e}_\theta$
 $\dot{\hat{e}}_\theta = (-\sin\theta \hat{i} + \cos\theta \hat{j})\dot{\theta} = \dot{\theta} \hat{e}_\lambda$

FBD of mass,

$\vec{T} = T \hat{e}_\lambda$
 $\vec{w} = -mg \hat{j}$

UMP: $\Sigma \vec{F} = m \vec{a}$

$\hat{j} = \cos\theta \hat{e}_\lambda + \sin\theta \hat{e}_\theta$
 $\Rightarrow \frac{T}{m} \hat{e}_\lambda + mg(-\hat{j}) = m(L\dot{\theta} \hat{e}_\theta + L\dot{\theta}^2 \hat{e}_\lambda - \ddot{\theta} \hat{e}_\lambda + \dot{\theta} \dot{\hat{e}}_\theta)$

dotting w/ $\hat{e}_\lambda$ and $\hat{e}_\theta$,

$\text{EoM} \Rightarrow \frac{T}{m} - g \cos\theta = L\dot{\theta}^2 - \ddot{\theta}$; $g \sin\theta = (L + \dot{L})\dot{\theta}$
 in Polar Coords.

If $g=0$, $\hat{e}_\theta$ component of acceleration will be $0 = (L + \dot{L})\dot{\theta}$
 $\Rightarrow \dot{\theta} = 0$
 $\therefore$ no ω in $\hat{e}_\theta$, $\therefore$ Motion is purely radial.

Explanation of why radial motion above

1.2.4 $L_0 = 0$

when $L_0 = 0$,

$$T = K(L - L_0) = KL$$

Using EoM

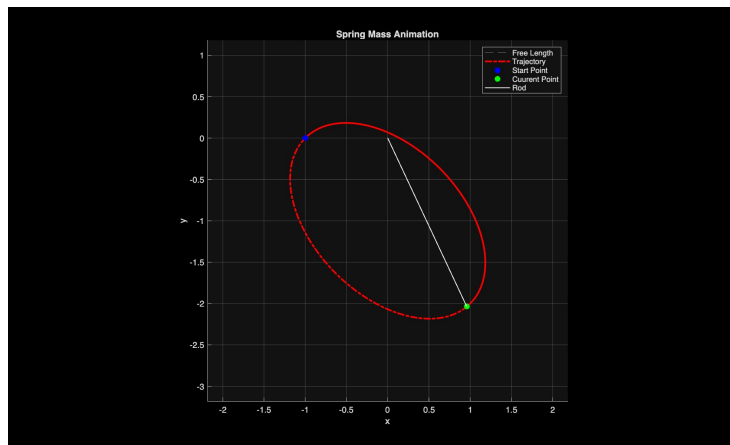
$$\boxed{-\frac{T}{L}x = m\ddot{x} \quad \text{and} \quad -\frac{T}{L}y - mg = m\ddot{y}}$$

$$a_x = -\frac{KL}{mK}x = -\frac{K}{m}x = \text{const.}$$

$$a_y = -\frac{KL}{mK}y - g = -\frac{K}{m}y - g = \text{const.}$$

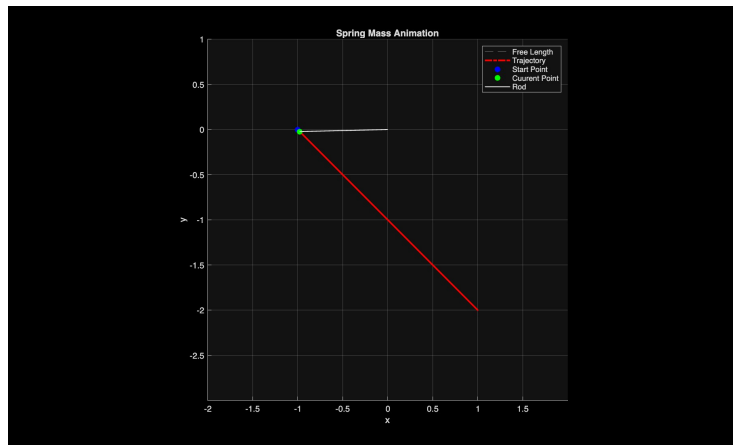
for const $\vec{a}$, $\vec{v}$ should be linear, $\vec{r}$ should be parabolic.

From the above explanation, we might expect the motion to be parabolic or elliptical as acceleration is constant.



Elliptical Motion

If v_x and v_y were both 0 at the initial state, then as acceleration and velocity would be constant, we would expect the motion to be a straight line.



$$v_0 = 0 \text{ and } L_0 = 0$$

I did notice that for the given parameters I had selected for the system:

```
% System Constants
p.m = 1; % Mass of bob (kg)
p.g = 10; % m/s^2
p.k = 10; % Spring coef (N/m)
p.l0 = 0; % Free Length of Spring (m)
```

When $[x_0, y_0] = [1, 0]$ and $v_x = v_y$, the motion is straight line motion as opposed to elliptical.

